Data Analytics Task - A/B Testing and SQL

This notebook is an A/B testing analysis project that involves data cleaning, SQL operations, and statistical testing to compare the performance of two groups (Test and Control) based on their click-through rates (CTR) for a specific banner.

Import Libraries

In [14]:
```
import pandas as pd
from scipy.stats import ttest_ind, chi2_contingency
from sqlalchemy import create_engine, text
import matplotlib.pyplot as plt
```

Load Data

In [15]:
```
test_customer_keys = pd.read_csv('test_customer_keys.csv')
banner_views = pd.read_csv('banner_views.csv')
banner_clicks = pd.read_csv('banner_clicks.csv')
```

Three CSV files are loaded into Pandas DataFrames:

- test_customer_keys: Contains customer keys and their variant (Test or Control).
- banner_views: Records of banner views, including customer keys, banner names, and timestamps.
- banner_clicks: Records of banner clicks, including customer keys, banner names, and timestamps.

In [29]:
```
# Display first 5 rows of 'test customer keys' table
test_customer_keys.head()
```

Out[29]:

	customer_key	customer_variant
0	a8cf21a461205975a97790c5aa05501a	Test
1	c05cca21bfaac87d226beb5b5a0f5689	Test
2	18e47b847aa6e50354a1ad3d60d4604d	Test
3	ecf77c568107ba2bda40360e0973efb0	Test
4	4fa8b68bb88c504a60d267225a591ae8	Test

In [17]:
```
# Display first 5 rows of 'test customer keys' table
banner_views.head()
```

Out[17]:

	customer_key	banner_name	created_at	
0	3b2f32d81bf642ca63a331c87ed5ce58	BFCbeans	2024-10-31 0:09:51	7
1	b6928adde107030b9f57f2ca7feb4506	Burstbybreadfast	2024-10-31 0:40:34	f
2	3d148ae720c687b74b377a43db378373	Brekkienewdesign	2024-10-31 12:25:19	5
3	c3dad81df9224d24c93d558241e2cdd7	Breakfastbasics	2024-10-31 3:37:55	f
4	f4a585510b82bee129de358949f5a6c1	Halwani	2024-10-31 1:09:03	0

In [18]:
```
banner_clicks.head()
```

Out[18]:

	customer_key	banner_name	create
0	19881d50dacce269056073ba0fcebb42	lunchbox2024	2024-31 2:1
1	8de3265c7a688f1d8cf80a3899ce5e6e	635a8n16277e3d0k0e69e89r	2024-31 9:0
2	NaN	635a6cfa90c491000edf413e	2024-01 4:5
3	5bf181ccb427a03e38414bb53c8efc54	fd66FreeDelivery	2024-31 0:1
4	28db8e97291ba71752097fa71599f90b	Pril	2024-01 23:33:

Data Cleaning

In [19]:
```
# Check for missing values
print("Missing values in test_customer_keys:")
print(test_customer_keys.isnull().sum())

print("\nMissing values in banner_views:")
print(banner_views.isnull().sum())

print("\nMissing values in banner_clicks:")
print(banner_clicks.isnull().sum())
```

```
Missing values in test_customer_keys:
customer_key        0
customer_variant    0
dtype: int64

Missing values in banner_views:
customer_key     453
banner_name        0
created_at         0
session_key     4298
dtype: int64

Missing values in banner_clicks:
customer_key    1111
banner_name        0
created_at         0
session_key     2308
dtype: int64
```

- Missing values are found in banner views and banner clicks for customer key and session_key.

In [20]:
```
# Drop rows with missing values (if necessary)
test_customer_keys.dropna(inplace=True)
banner_views.dropna(inplace=True)
banner_clicks.dropna(inplace=True)

# Verify no missing values remain
print("Missing values after cleaning:")
print(test_customer_keys.isnull().sum())
print(banner_views.isnull().sum())
print(banner_clicks.isnull().sum())
```

```
Missing values after cleaning:
customer_key        0
customer_variant    0
dtype: int64
customer_key    0
banner_name     0
created_at      0
session_key     0
dtype: int64
customer_key    0
banner_name     0
created_at      0
session_key     0
dtype: int64
```

- Rows with missing values are dropped from all datasets.

Create SQLite Database

In [21]:
```
# Create an in-memory SQLite database
engine = create_engine('sqlite:///:memory:')

# Load DataFrames into SQL tables
test_customer_keys.to_sql('test_customer_keys', engine, index=False)
banner_views.to_sql('banner_views', engine, index=False)
banner_clicks.to_sql('banner_clicks', engine, index=False)
```

Out[21]: 13579

Analysis

Create Customer Groups View

In [22]:
```
# Drop the view if it already exists
drop_query = text("DROP VIEW IF EXISTS customer_groups;")

# Create the view using sqlalchemy
create_query = text("""
CREATE VIEW customer_groups AS
SELECT c.customer_key,
       CASE WHEN t.customer_key IS NOT NULL THEN 'Test' ELSE 'Control' END
AS customer_group
FROM (SELECT DISTINCT customer_key FROM banner_views UNION SELECT DISTINCT
customer_key FROM banner_clicks) c
LEFT JOIN test_customer_keys t ON c.customer_key = t.customer_key;
""")

# Execute the queries
with engine.connect() as connection:
    connection.execute(drop_query)  # Drop the view if it exists
    connection.execute(create_query)  # Create the view
    connection.commit()  # save the transaction
```

In [23]:
```
# Check the first 5 rows of the view
customer_groups_sample = pd.read_sql("SELECT * FROM customer_groups;",
engine)

customer_groups_sample
```

Out[23]:

	customer_key	customer_group
0	0004c5e4353e6594b231eb346cf79d23	Control
1	000528824f9c6c395c47314ec75edfbe	Control
2	0007b33129fae8cfecbdd697cfdeb782	Test
3	0008400aabbd30af11e692bdac97a4c5	Control
4	001062a0e983a1d23c651bda00eee0e0	Control
...	...	...
15435	ffc9b947df7f09ea8d2e4e2aef2865a1	Control
15436	ffd4ef017c8d3399937c783bc3119715	Control
15437	ffe032f875593c10cad1e6886eae0c3c	Control
15438	ffe802b630ab228bdd90bfa6eb61b994	Control
15439	fffb21584fa169dd5a54ccd26e1e5673	Control

15440 rows × 2 columns

- The view assigns each customer to either the "Test" or "Control" group based on whether their key exists in the test_customer_keys table.

Calculate Impressions and Clicks

In [ ]:

```
# Calculate impressions and clicks for banner
query = """
SELECT cg.customer_group,
       COUNT(bv.created_at) AS impressions,
       COUNT(bc.created_at) AS clicks
FROM customer_groups cg
LEFT JOIN banner_views bv ON cg.customer_key = bv.customer_key AND
bv.banner_name = 'Juhayna60'
LEFT JOIN banner_clicks bc ON cg.customer_key = bc.customer_key AND
bc.banner_name = 'Juhayna60'
GROUP BY cg.customer_group;
"""

# Execute the query
results = pd.read_sql(query, engine)
print("Impressions and Clicks:")
print(results)
```

```
Impressions and Clicks:
  customer_group  impressions  clicks
0        Control         5991    1006
1           Test         1070     201
```

- The query calculates the number of impressions (views) and clicks for the banner "Juhayna60" for both Test and Control groups.

In [25]:

```
# Calculate CTR (Click-Through Rate)
results['CTR'] = (results['clicks'] / results['impressions']) * 100
print("CTR Results:")
print(results)
```

```
CTR Results:
  customer_group  impressions  clicks        CTR
0        Control         5991    1006  16.791854
1           Test         1070     201  18.785047
```

- The results show that the Test group has a slightly higher CTR (18.79%) compared to the Control group (16.79%).

In [26]:
```
# Statistical test
contingency_table = [
    [results.loc[results['customer_group'] == 'Test', 'clicks'].values[0],
     results.loc[results['customer_group'] == 'Test',
'impressions'].values[0] - results.loc[results['customer_group'] == 'Test',
'clicks'].values[0]],
    [results.loc[results['customer_group'] == 'Control',
'clicks'].values[0],
     results.loc[results['customer_group'] == 'Control',
'impressions'].values[0] - results.loc[results['customer_group'] ==
'Control', 'clicks'].values[0]]
]

# Perform Chi-Square Test
chi2, p, dof, expected = chi2_contingency(contingency_table)
print(f"p-value: {p}")
```

```
p-value: 0.12084780876355133
```

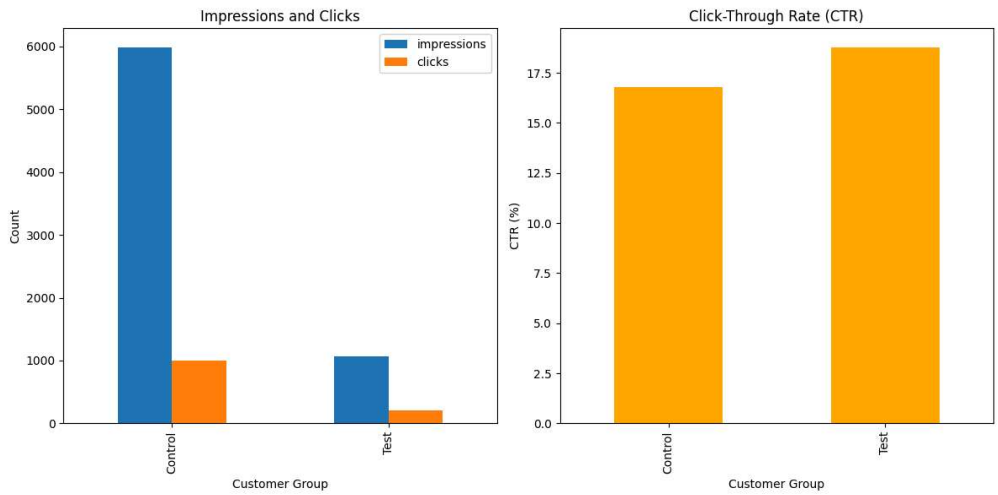
- A chi-square test is performed to determine if the difference in CTR between the Test and Control groups is statistically significant.

- The p-value is 0.1208, which is greater than 0.05, indicating that the difference is not statistically significant.

In [27]:
```
# Visualizing the results
fig, ax = plt.subplots(1, 2, figsize=(12, 6))
# Bar plot for impressions and clicks
results.set_index('customer_group')[['impressions',
'clicks']].plot(kind='bar', ax=ax[0])
ax[0].set_title('Impressions and Clicks')
ax[0].set_ylabel('Count')
ax[0].set_xlabel('Customer Group')

# CTR plot
results.set_index('customer_group')['CTR'].plot(kind='bar', ax=ax[1],
color='orange')
ax[1].set_title('Click-Through Rate (CTR)')
ax[1].set_ylabel('CTR (%)')
ax[1].set_xlabel('Customer Group')

plt.tight_layout()
plt.show()
```



Interpretation:

- The Test group has a slightly higher CTR than the Control group.

- However, the difference is not statistically significant (p-value > 0.05), meaning the observed difference could be due to random variation.

- Since the difference in CTR is not statistically significant, it is not advisable to implement the new banner weights.

Exported with runcell — convert notebooks to HTML or PDF anytime at runcell.dev.